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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Capstone Project

**Automatic Bone Fracture Detection and Classification from
X-Ray Images Using Deep Learning with VGG16 and
ResNet50**

Team no. 6

MLA0405 – Deep Learning

Guided By

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Abstract

- Develops an automated system for detecting and classifying bone fractures from X-ray images using Deep Learning.
- Utilizes Transfer Learning with VGG16 and ResNet50 models to extract features and improve classification accuracy.
- Performs data preprocessing, model training, and fracture classification on X-ray image datasets.
- Evaluates and compares both models using performance metrics such as Accuracy, Precision, Recall, and F1-Score.
- Provides fast and accurate fracture predictions to assist radiologists in improving diagnosis and clinical decision-making.

Aim and Objective

Aim

To develop an automated deep learning system for detecting and classifying bone fractures from X-ray images using **VGG16** and **ResNet50** transfer learning models, enabling accurate and efficient diagnosis.

Objectives

- Collect and preprocess bone X-ray images to create a high-quality dataset for model training.
- Extract meaningful image features using transfer learning with VGG16 and ResNet50.
- Detect and classify bone fractures accurately using deep learning techniques.
- Evaluate and compare the performance of both models using metrics such as Accuracy, Precision, Recall, and F1-Score.
- Develop a prediction system that provides fast and reliable fracture detection results to assist healthcare professionals.

Introduction

- Bone fractures are common injuries that require accurate and timely diagnosis for effective treatment.
- X-ray imaging is the standard method used by doctors to identify bone fractures.
- Manual analysis of X-ray images can be time-consuming and may result in diagnostic errors.
- This project uses Deep Learning with VGG16 and ResNet50 transfer learning models to automatically detect and classify bone fractures.
- The proposed system aims to improve diagnostic accuracy, reduce analysis time, and assist healthcare professionals in making faster clinical decisions.

Literature Review

S.No	Year of Publication	Author	Title of the Journal / Book	Methodology Used	Algorithm Used	Key Findings
1	2023	Olczak et al.	Artificial Intelligence for Fracture Detection in Musculoskeletal Radiographs	Deep Learning-based Medical Image Analysis	CNN	AI-assisted diagnosis improved fracture detection accuracy and reduced diagnostic errors.
2	2022	Li et al.	Bone Fracture Classification Using Transfer Learning	Transfer Learning on X-ray Images	VGG16	VGG16 achieved high classification accuracy with reduced training time.
3	2022	Cheng et al.	Automatic Bone Fracture Detection Using Deep Residual Networks	Feature Extraction and Image Classification	ResNet50	ResNet50 provided better feature extraction and higher accuracy than conventional CNNs.
4	2021	Kumar et al.	Deep Learning-Based Bone Fracture Detection from X-Ray Images	Image Preprocessing and CNN Classification	VGG16 & ResNet50	Comparative analysis showed ResNet50 outperformed VGG16 in fracture detection.

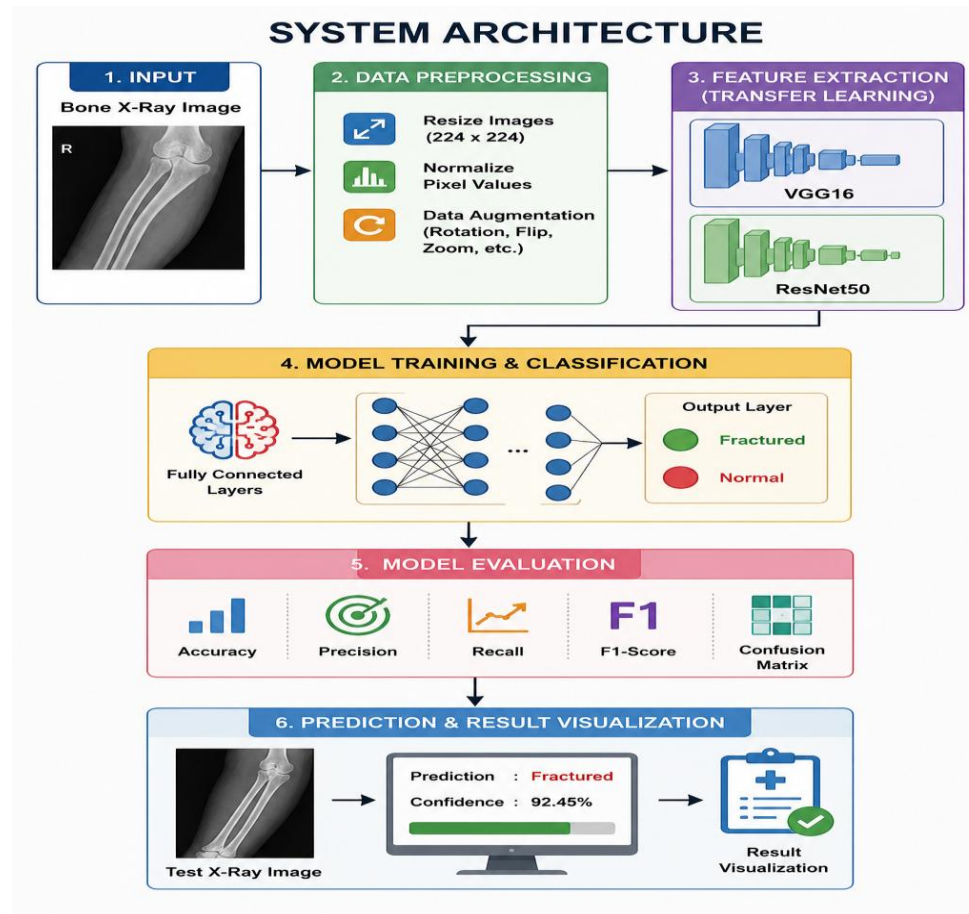
Problem Statement

- Manual interpretation of X-ray images is time-consuming.
- Diagnosis depends heavily on the experience of radiologists.
- Small fractures may be overlooked.
- Hospitals receive thousands of X-rays daily.
- There is a need for a fast, accurate, and automated fracture detection system.

Existing System vs Proposed System

Existing System	Proposed System
Bone fractures are diagnosed manually by radiologists through visual inspection of X-ray images.	Bone fractures are detected and classified automatically using Deep Learning models (VGG16 and ResNet50).
Diagnosis is time-consuming and depends on the expertise of the radiologist.	Provides fast and consistent predictions with minimal human intervention.
Manual analysis may lead to diagnostic errors, especially for subtle fractures.	Transfer Learning improves feature extraction and enhances detection accuracy.
Traditional methods have limited capability to handle large volumes of X-ray images efficiently.	The system efficiently processes large datasets and supports real-time fracture detection.
No automated comparison of different deep learning models for optimal performance.	Compares the performance of VGG16 and ResNet50 using evaluation metrics to identify the best-performing model.

Overall System Architectural Design





Sustainable Development Goal (SDG) Implementation



SDG Goal	Implementation in the Project
SDG 3 – Good Health and Well-being	The proposed system assists doctors and radiologists in accurately detecting and classifying bone fractures, leading to faster diagnosis and improved patient care.
SDG 9 – Industry, Innovation and Infrastructure	Utilizes advanced Deep Learning techniques (VGG16 and ResNet50) to develop an innovative AI-based healthcare solution for medical image analysis.
SDG 10 – Reduced Inequalities	Supports healthcare facilities with limited access to experienced radiologists by providing automated fracture detection, improving access to quality healthcare.
SDG 17 – Partnerships for the Goals	Encourages collaboration between healthcare professionals, AI researchers, and technology developers to enhance medical diagnostic systems.

Proposed Methodology and Algorithm

Proposed Methodology

1. Collect and preprocess bone X-ray images.
2. Extract image features using **VGG16** and **ResNet50**.
3. Train the models for fracture detection and classification.
4. Evaluate model performance using standard metrics.
5. Predict fractures from new X-ray images and display the results.

Algorithms Used

- **VGG16** – Extracts image features and classifies bone fractures.
- **ResNet50** – Deep residual network for accurate fracture detection.
- **Transfer Learning** – Uses pre-trained models to improve accuracy and reduce training time.
- **Softmax Classifier** – Predicts whether the X-ray is Fractured or Normal.



Module Planning

Module No.	Module Name	Description
Module 1	Data Collection and Preprocessing	Collect bone X-ray images, preprocess them, and prepare the dataset for training.
Module 2	Feature Extraction using VGG16 & ResNet50	Extract important image features using transfer learning with VGG16 and ResNet50.
Module 3	Bone Fracture Detection and Classification	Train the models to detect and classify bone fractures from X-ray images.
Module 4	Model Evaluation and Performance Comparison	Evaluate and compare the performance of both models using accuracy and other metrics.
Module 5	Prediction and Result Visualization	Predict fractures on new X-ray images and display the results with confidence scores.

Conclusion

- The proposed system successfully automates bone fracture detection and classification from X-ray images using Deep Learning.
- Transfer Learning with **VGG16** and **ResNet50** enables accurate and efficient feature extraction.
- Performance comparison helps identify the model with better fracture detection accuracy.
- The system reduces diagnosis time and assists radiologists in making faster and more reliable clinical decisions.
- This AI-based solution has the potential to improve healthcare services and support future advancements in medical image analysis.

References

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